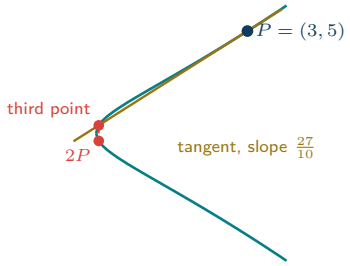
PROJECTIVE GEOMETRY LADDER 8 OF 9

Cubics: chords, tangents and Bézout

A line meets a cubic in three points, so two rational points force a third, and the points form a group.

1 / THE RULE AND THE PICTURE

A line meets a cubic in 3 points counted with multiplicity, so two rational points give a third



$y^2 = x^3 - 2$: the tangent at P meets the curve again, and reflecting gives $2P$

Substituting a line $y = \lambda x + \nu$ into the cubic gives a cubic equation in x whose three roots add to λ^2, so if two roots are rational so is the third. Bézout guarantees the three, counting a tangent as two:

$$x_1 + x_2 + x_3 = \lambda^2, \qquad P + Q = \text{reflection of the third point}$$

The reflection makes the operation commutative, with the point at infinity $O = (0 : 1 : 0)$ as the zero.

2 / DOUBLING AND TRIPLING A POINT

The coordinates stay rational and grow quickly

step	computation	check
P	$(3, 5)$, and $5^2 = 27 - 2$	$25 = 25$
tangent at P	slope $\frac{3x^2}{2y} = \frac{27}{10}$	$\lambda = \frac{27}{10}$
third point of the line	$x = \lambda^2 - 2 \cdot 3 = \frac{129}{100}$, $y = 5 + \lambda(\frac{129}{100} - 3) = \frac{383}{1000}$	on the curve
$2P$	reflect in the x-axis: $\left(\frac{129}{100}, -\frac{383}{1000}\right)$	$y^2 = x^3 - 2$ holds
$3P = P + 2P$	chord through P and $2P$: $\left(\frac{164323}{29241}, -\frac{66234835}{5000211}\right)$	on the curve

3 / THE POINT AT INFINITY

A flex, so the zero is its own negative

In homogeneous form the curve is $Y^2Z = X^3 - 2Z^3$, and $Z = 0$ forces $X^3 = 0$, so $(0 : 1 : 0)$ is a triple point of the intersection with the line at infinity and a flex of the curve.
The tangent there meets the curve three times at O itself, which says that $O + O = O$, and the line through P and its reflection $-P$ meets the curve a third time at O, which says $P + (-P) = O$.

4 / CONICS AGAINST CUBICS

A parametrisation against a group

On a conic one rational point and the line trick of the previous sheet give every rational point, while on a cubic the chord and tangent method only generates new points from old ones, and the point $(3, 5)$ generates infinitely many rational points $P, 2P, 3P, \ldots$ with ever larger numerators and denominators.
Mordell proved in 1922 that the rational points of such a curve are generated by finitely many of them, which is the beginning of the theory of elliptic curves.

5 / TRY THESE

Check by hand

(a) Verify that $2P = (\frac{129}{100}, -\frac{383}{1000})$ lies on the curve: $x^3 - 2 = 0.146689$ and $y^2 = 0.146689$.
(b) Find the tangent slope at $(3, 5)$: differentiating $y^2 = x^3 - 2$ gives $2y\,y' = 3x^2$, so $y' = \frac{27}{10}$.
(c) The roots of the cubic from the tangent are $3, 3, \frac{129}{100}$, which add to $\frac{729}{100} = \lambda^2$.

ASIDE / WHY THREE

The count 3 is Bézout for a line and a cubic, and it is the reason the cubic has a group law, while the count 2 for a conic gives only a parametrisation.

LINKS TO OTHER SHEETS

The Bézout sheet proves the count; the algebra ladder explains why this addition is associative with an identity and inverses, the group axioms.

Convention. λ is the slope of the chord or tangent; $O = (0 : 1 : 0)$ is the point at infinity; $P + Q$ is the reflection of the third intersection point; computations use exact fractions.